

Pinned group summary: $SU(n=6, q=3, \text{eps}=-1)$

Pinning-Sympy automated output

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1 Basic attributes

Group	SU(n=6, q=3, eps=-1)
Matrix size	6×6
Rank	3

1.1 Skew-hermitian form

Dimension	6
Witt index	3
Primitive element	$\sqrt{d}$
Epsilon	-1

Matrix:
$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \end{bmatrix}$$

1.2 Generic Lie algebra element

$$\begin{bmatrix} \sqrt{d}x_{i00} + x_{r00} & \sqrt{d}x_{i01} + x_{r01} & \sqrt{d}x_{i02} + x_{r02} & -x_{r03} & -\sqrt{d}x_{i13} + x_{r13} & -\sqrt{d}x_{i23} + x_{r23} \\ \sqrt{d}x_{i10} + x_{r10} & \sqrt{d}x_{i11} + x_{r11} & \sqrt{d}x_{i12} + x_{r12} & \sqrt{d}x_{i13} + x_{r13} & -x_{r14} & -\sqrt{d}x_{i24} + x_{r24} \\ \sqrt{d}x_{i20} + x_{r20} & \sqrt{d}x_{i21} + x_{r21} & -\sqrt{d}(x_{i00} + x_{i11}) + x_{r22} & \sqrt{d}x_{i23} + x_{r23} & \sqrt{d}x_{i24} + x_{r24} & -x_{r25} \\ -x_{r30} & -\sqrt{d}x_{i40} + x_{r40} & -\sqrt{d}x_{i50} + x_{r50} & \sqrt{d}x_{i00} - x_{r00} & \sqrt{d}x_{i10} - x_{r10} & \sqrt{d}x_{i20} - x_{r20} \\ \sqrt{d}x_{i40} + x_{r40} & -x_{r41} & -\sqrt{d}x_{i51} + x_{r51} & \sqrt{d}x_{i01} - x_{r01} & \sqrt{d}x_{i11} - x_{r11} & \sqrt{d}x_{i21} - x_{r21} \\ \sqrt{d}x_{i50} + x_{r50} & \sqrt{d}x_{i51} + x_{r51} & -x_{r52} & \sqrt{d}x_{i02} - x_{r02} & \sqrt{d}x_{i12} - x_{r12} & -\sqrt{d}(x_{i00} + x_{i11}) - x_{r22} \end{bmatrix}$$

1.3 Generic torus element

$$\begin{bmatrix} t_0 & 0 & 0 & 0 & 0 & 0 \\ 0 & t_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & t_2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{t_0} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{t_1} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{t_2} \end{bmatrix}$$

1.4 Trivial characters

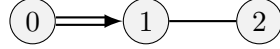
$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

2 Root system

Summary Properties

Dynkin type	C3
Reduced	True
Simply laced	False
Number of roots	18

2.1 Dynkin diagram



2.2 Cartan matrix

$$\begin{bmatrix} 2 & -2 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

2.3 Roots and coroots

Root	Coroot	Simple	Sign	Multipliable
(-2, 0, 0, 0, 0, 0)	(-1, 0, 0, 1, 0, 0)	No	-	No
(-1, -1, 0, 0, 0, 0)	(-1, -1, 0, 1, 1, 0)	No	-	No
(-1, 0, -1, 0, 0, 0)	(-1, 0, -1, 1, 0, 1)	No	-	No
(-1, 0, 1, 0, 0, 0)	(-1, 0, 1, 1, 0, -1)	No	-	No
(-1, 1, 0, 0, 0, 0)	(-1, 1, 0, 1, -1, 0)	No	-	No
(0, -2, 0, 0, 0, 0)	(0, -1, 0, 0, 1, 0)	No	-	No
(0, -1, -1, 0, 0, 0)	(0, -1, -1, 0, 1, 1)	No	-	No
(0, -1, 1, 0, 0, 0)	(0, -1, 1, 0, 1, -1)	No	-	No
(0, 0, -2, 0, 0, 0)	(0, 0, -1, 0, 0, 1)	No	-	No
(0, 0, 2, 0, 0, 0)	(0, 0, 1, 0, 0, -1)	Yes	+	No
(0, 1, -1, 0, 0, 0)	(0, 1, -1, 0, -1, 1)	Yes	+	No
(0, 1, 1, 0, 0, 0)	(0, 1, 1, 0, -1, -1)	No	+	No
(0, 2, 0, 0, 0, 0)	(0, 1, 0, 0, -1, 0)	No	+	No
(1, -1, 0, 0, 0, 0)	(1, -1, 0, -1, 1, 0)	Yes	+	No
(1, 0, -1, 0, 0, 0)	(1, 0, -1, -1, 0, 1)	No	+	No
(1, 0, 1, 0, 0, 0)	(1, 0, 1, -1, 0, -1)	No	+	No
(1, 1, 0, 0, 0, 0)	(1, 1, 0, -1, -1, 0)	No	+	No
(2, 0, 0, 0, 0, 0)	(1, 0, 0, -1, 0, 0)	No	+	No

2.4 Linear combinations of roots

Definition 2.1. For two roots α, β in a root system Φ , the set of positive integral linear combinations of α, β that are also roots is denoted

$$(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$$

α	β	Pairs (i, j)
$(-2, 0, 0, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$(1, 1), (1, 2)$
$(-2, 0, 0, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$(1, 1), (1, 2)$
$(-2, 0, 0, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$(1, 1), (1, 2)$
$(-2, 0, 0, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$(1, 1), (1, 2)$
$(-1, -1, 0, 0, 0, 0)$	$(-1, 1, 0, 0, 0, 0)$	$(1, 1)$
$(-1, -1, 0, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	$(1, 1)$
$(-1, -1, 0, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$(1, 1)$
$(-1, -1, 0, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	$(1, 1), (2, 1)$
$(-1, -1, 0, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$(1, 1)$
$(-1, -1, 0, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$(1, 1)$
$(-1, -1, 0, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$(1, 1)$
$(-1, -1, 0, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	$(1, 1), (2, 1)$
$(-1, 0, -1, 0, 0, 0)$	$(-1, 0, 1, 0, 0, 0)$	$(1, 1)$
$(-1, 0, -1, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	$(1, 1)$
$(-1, 0, -1, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	$(1, 1), (2, 1)$
$(-1, 0, -1, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$(1, 1)$
$(-1, 0, -1, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$(1, 1)$
$(-1, 0, -1, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$(1, 1)$
$(-1, 0, -1, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$(1, 1)$
$(-1, 0, -1, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	$(1, 1), (2, 1)$
$(-1, 0, 1, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	$(1, 1)$
$(-1, 0, 1, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	$(1, 1), (2, 1)$
$(-1, 0, 1, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	$(1, 1)$
$(-1, 0, 1, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$(1, 1)$
$(-1, 0, 1, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$(1, 1)$
$(-1, 0, 1, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$(1, 1)$
$(-1, 0, 1, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	$(1, 1), (2, 1)$
$(-1, 1, 0, 0, 0, 0)$	$(0, -2, 0, 0, 0, 0)$	$(1, 1), (2, 1)$
$(-1, 1, 0, 0, 0, 0)$	$(0, -1, -1, 0, 0, 0)$	$(1, 1)$
$(-1, 1, 0, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	$(1, 1)$
$(-1, 1, 0, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$(1, 1)$
$(-1, 1, 0, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$(1, 1)$
$(-1, 1, 0, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$(1, 1)$
$(-1, 1, 0, 0, 0, 0)$	$(2, 0, 0, 0, 0, 0)$	$(1, 1), (2, 1)$
$(0, -2, 0, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	$(1, 1), (1, 2)$
$(0, -2, 0, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$(1, 1), (1, 2)$
$(0, -2, 0, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$(1, 1), (1, 2)$
$(0, -1, -1, 0, 0, 0)$	$(0, -1, 1, 0, 0, 0)$	$(1, 1)$
$(0, -1, -1, 0, 0, 0)$	$(0, 0, 2, 0, 0, 0)$	$(1, 1), (2, 1)$
$(0, -1, -1, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	$(1, 1)$
$(0, -1, -1, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	$(1, 1), (2, 1)$
$(0, -1, -1, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$(1, 1)$
$(0, -1, -1, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$(1, 1)$
$(0, -1, 1, 0, 0, 0)$	$(0, 0, -2, 0, 0, 0)$	$(1, 1), (2, 1)$
$(0, -1, 1, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$(1, 1)$
$(0, -1, 1, 0, 0, 0)$	$(0, 2, 0, 0, 0, 0)$	$(1, 1), (2, 1)$
$(0, -1, 1, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$(1, 1)$
$(0, -1, 1, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$(1, 1)$
$(0, 0, -2, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$(1, 1), (1, 2)$
$(0, 0, -2, 0, 0, 0)$	$(1, 0, 1, 0, 0, 0)$	$(1, 1), (1, 2)$
$(0, 0, 2, 0, 0, 0)$	$(0, 1, -1, 0, 0, 0)$	$(1, 1), (1, 2)$
$(0, 0, 2, 0, 0, 0)$	$(1, 0, -1, 0, 0, 0)$	$(1, 1), (1, 2)$
$(0, 1, -1, 0, 0, 0)$	$(0, 1, 1, 0, 0, 0)$	$(1, 1)$